

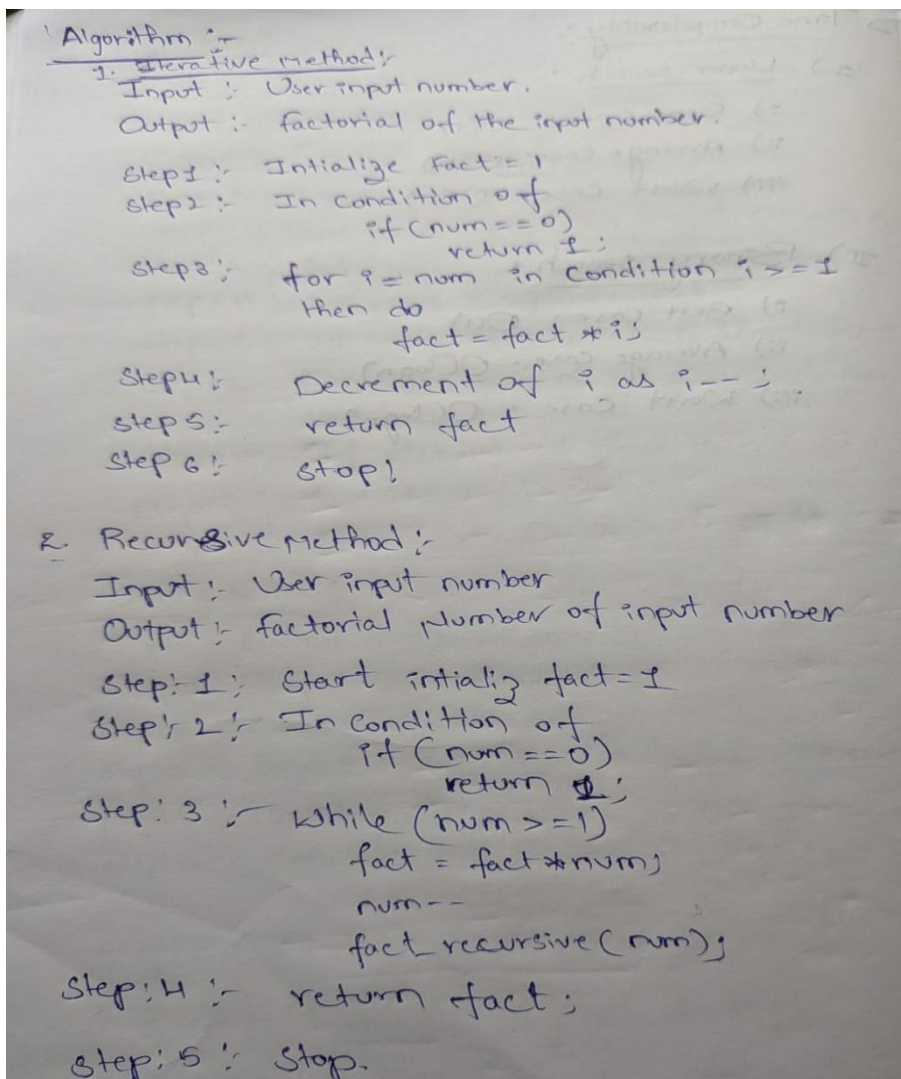
Practical: 4

Implementation and Time analysis of factorial program using iterative and recursive method

18/08/2026

Aim: To Implement and Time analysis of factorial program using iterative and recursive method.

Algorithm:



Algorithm :-

1. Iterative Method:-
Input :- User input number.
Output :- factorial of the input number.
Step 1 :- Initialize fact = 1
Step 2 :- In condition of
if (num == 0)
return 1;
Step 3 :- for i = num in condition i >= 1
then do
fact = fact * i;
Step 4 :- Decrement of i as i--;
Step 5 :- return fact
Step 6 :- stop!

2. Recursive method:-
Input :- User input number
Output :- factorial number of input number
Step 1 :- Start initializ fact = 1
Step 2 :- In condition of
if (num == 0)
return 1;
Step 3 :- while (num >= 1)
fact = fact * num;
num--
fact recursive (num);
Step 4 :- return fact;
Step 5 :- Stop.

Program:

factorial program using iterative and recursive method

```
#include<iostream>
using namespace std;
```

```
int fact_iterative(int num){
    int fact = 1;
    if(num == 0)
        return 1;
    for(int i=num; i>=1; i--){
        fact = fact*i;
    }
    return fact;
}

int fact_recursive(int num){
    int fact = 1;
    if(num == 0)
        return 1;
    while(num >= 1){
        fact = fact*num;
        num--;
        fact_recursive(num);
    }
    return fact;
}

int main(){
    cout<<"-----Factorial-----"<<endl;
    int n;
    cout<<"Enter your number: ";
    cin>>n;
    int i = fact_iterative(n);

    cout<<"Iterative method:"<<endl;
    cout<<n<<" factorial is "<<i;
    cout<<endl<<endl;
    int r = fact_recursive(n);
    cout<<"Recursive method:"<<endl;
    cout<<n<<" factorial is "<<r;

    return 0;
}
```

Output:

```

C:\Users\Chandu\OneDrive\D × + v
-----Factorial-----
Enter your number: 5
Iterative method:
5 factorial is 120

Recursive method:
5 factorial is 120
-----
Process exited after 2.87 seconds with return value 0
Press any key to continue . . .

```

Time Complexity:

Time Complexity :-

1. Iterative method :-

```

fact = 1;
if (num == 0) {
    return 1;
}
for (i = num; i >= 1; i--) {
    fact = fact * i;
}
return fact;

```

1
 1
 1
 $n + n + 1 + n$
 $2n + 2$
 n
 1

$2n + 2 + n$
 $3n + 2$
 n

The Time Complexity is $O(n)$

Note:-
Best case $O(1)$
Average $O(n)$
Worst $O(n)$
Sec $O(1)$

2) Recursive Method :-

```

int fact(int num) {
    int fact = 1;
    if (num == 0) {
        return 1;
    }
    while (num >= 1) {
        fact = fact * num;
        num--;
    }
    fact(num);
    return fact;
}

```

1
 1
 1
 1
 n
 n
 n
 n
 1

$3n + 4$
 n

Time Complexity is $O(n)$
Space Complexity is $O(1)$

Note
Best case $O(n)$
Average case $O(n)$
Worst case $O(n)$

Conclusion:

The implementation of this program and the execution is successfully got executed and the time complexity in the Three cases are $O(n)$ in the both Iterative, recursive methods And Space complexity of both methods are $O(1)$.